

# Louay Guetat

## AI ENGINEER

Tunis, Tunisia · Open to relocation worldwide

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*AI engineer with a strong full-stack background. I work mainly on multi-agent systems, LLM integration and RAG pipelines, and I like building things all the way through, from the retrieval and agent logic to the APIs and UI that ship them. I care a lot about evaluation, guardrails and keeping model outputs grounded in real sources. Right now I'm focused on financial research agents and an AI diligence platform for private markets.*

## WORK EXPERIENCE

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### AI Engineer · Alpha Momentum

Remote · US Startup | February 2025 – Present

Led AI/ML work on a financial research platform built around multi-agent systems and RAG pipelines, and looked after the full-stack side too (React web, React Native mobile, and a serverless Node.js API). Since August 2026 I've also been leading Capital Thread, an AI diligence platform for private markets (covered under Selected Projects), through Alpha Momentum.

- Architected a multi-agent system with LangChain and LangGraph, including specialized agents for equity research, valuation, portfolio optimization, risk, technical analysis and market sentiment. It brings together several LLM providers (OpenAI, Anthropic, Google Gemini, xAI, Groq, SambaNova, Ollama) with dynamic selection and fallback
- Developed Dexter, a financial research agent with custom tools for financial search, SEC filing analysis, web search (Tavily/Exa), and browser automation (Playwright), plus a markdown-based skills system for specialized analyses (DCF models, options) with frontmatter metadata
- Wrote an evaluation framework with rubric-based assessment, numerical-accuracy validation, tool-selection analysis, and confidence intervals, using LangSmith for tracing and performance monitoring across agent iterations
- Implemented a RAG pipeline with intelligent chunking, OpenAI embeddings, Neo4j graph storage for semantic relationships, and optimized retrieval over financial documents for context-aware responses
- Built FastAPI services for SEC filing processing (edgartools + custom parsers) extracting structured data from 10-K, 10-Q, and 8-K filings, with AWS Batch orchestration for 37k+ documents and incremental updates
- Wrote Python data-ingestion pipelines (BeautifulSoup, Selenium, aiohttp) for FMP, Federal Reserve, ECB, news, earnings transcripts, and market signals, with structured logging and error handling
- Built out the full-stack delivery layer: a React 18 + TypeScript web app with real-time AI streaming (SSE), a serverless Node.js API with LangChain integration, React Native apps, and GitHub Actions CI/CD to Docker/AWS

**Tech Stack:** Python (FastAPI, LangChain, LangGraph, LangSmith, Pandas, NumPy), OpenAI, Anthropic, Google Gemini, xAI, Groq, Neo4j, MongoDB, Redis, Docker, AWS (Lambda, ECS, ECR, Batch), edgartools, BeautifulSoup, Selenium, Playwright, TypeScript, React, Node.js

### Data Science Intern · Sagemcom

Tunis | February 2024 – August 2024

- Built multi-agent systems with LLM integration and embedding-based retrieval
- Built a full data pipeline: scraping and dataset assembly (Selenium, BeautifulSoup), retrieval with HuggingFace models and LangChain, and automated workflows with LangGraph
- Developed a ReactJS + Flask web interface for model interactions over a REST API

**Technologies:** Python, Flask, LangChain, LangGraph, HuggingFace, ReactJS, SQLite, REST API

## Full Stack Developer · WeTekup

Tunis | March 2025 – September 2025

- Built responsive React websites (SCSS) with Node.js/Express REST APIs over PostgreSQL
- Deployed to production on Hostinger with an Nginx reverse proxy, SSL/TLS, and performance tuning

**Technologies:** React, Node.js, Express.js, SCSS, PostgreSQL, Nginx

## Full Stack Intern · WeTekup (Khedma.site)

Tunis | July 2023 – September 2023

- Built a job-marketplace platform with a ReactJS front end and Django REST APIs over PostgreSQL

**Technologies:** ReactJS, Django, Python, PostgreSQL

## SELECTED PROJECTS

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### Capital Thread · AI Diligence Platform for Private Markets · Client engagement via Alpha Momentum · Aug 2026 – Present

One-click diligence pipeline for private credit, PE and venture. It sources deals from SEC EDGAR and Form D, parses Inline XBRL, and generates financial models, DCF valuations and IC memos where every figure links back to its source filing. The LLMs (OpenAI, Anthropic) handle judgment and the write-up, while all the arithmetic runs in deterministic Python, so there are no hallucinated numbers. 437 unit tests cover the financial engines.

**Tech:** OpenAI API, Anthropic API, FastAPI, PostgreSQL, SQLAlchemy, SEC EDGAR, Arelle XBRL, React 19, TypeScript

### Alpha Momentum · Multi-Agent Financial Research · *alphamomentum.ai* · Live

Five LangGraph research workflows and a RAG pipeline (OpenAI embeddings + Neo4j) over a 37k+ document SEC corpus, delivered across web and mobile.

**Tech:** LangChain, LangGraph, OpenAI, Anthropic, Neo4j, FastAPI, AWS Batch

### Intellectia · Vision-Powered Math Learning · Team project

Math learning platform for children that turns text problems into visual representations using vision models, with interactive quizzes and performance dashboards.

**Tech:** TensorFlow, HuggingFace, React, Django, MongoDB, NumPy, Pandas

### PMBOK Risk Recommender · NLP / GNN · Team project

Chatbot-style project-risk recommender trained on PMBOK 11th-edition data, using a Graph Attention Network with a BERT embedder for text-based recommendations.

**Tech:** PyTorch, TensorFlow, BERT, GNN, NLP, Django, NumPy

### Yoga Pose Classifier · Computer Vision · Solo build

CNN-based yoga pose classification; a custom CNN reached 86% accuracy and ResNet reached 97% across multiple pose categories.

**Tech:** Python, CNN, ResNet, Django, NumPy, Pandas

## TECHNICAL SKILLS

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**AI/ML & LLM:** LangChain, LangGraph, LangSmith, OpenAI API, Anthropic API, Google Gemini, RAG, HuggingFace, multi-agent systems, prompt/agent evaluation

**Deep Learning:** TensorFlow, PyTorch, Keras, scikit-learn, CNN, ResNet, BERT, GNN, NLP, Computer Vision

**Data:** Pandas, NumPy, BeautifulSoup, Selenium, Playwright, aiohttp, SEC EDGAR, Arelle XBRL

**Backend:** Python (FastAPI, Flask, Django), Node.js (Express, Serverless)

**Databases:** PostgreSQL, MongoDB, Neo4j, MySQL, Redis, DynamoDB

**Cloud & DevOps:** AWS (Lambda, ECS, ECR, Batch, S3, CloudFront, Cognito), Docker, GitHub Actions

**Frontend:** React 19, TypeScript, React Native, Next.js, MUI, Tailwind CSS

## EDUCATION

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### Computer Engineering, Data Science · ESPRIT

*Tunis · 2021 – 2024*

- Thesis: “Generative AI with Business Knowledge” · Mention: Very Good

### Bachelor of Computer Science · ESSECT

*Tunis · 2018 – 2021*

- Foundation in software development, algorithms, and data structures

## ADDITIONAL INFORMATION

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- **Languages:** Arabic (Native), English (B2, Professional), French (B2, Professional)
- **Soft skills:** Problem-solving, collaboration, and clear communication